F FWallSim · Firewall Course

CLASS 11

SSL-VPN for remote workers

SSL-VPN settings, portals, ssl.root policy

Advanced | ~25 minutes | FWallSim v1.1.0 (FortiOS 7.4 style)

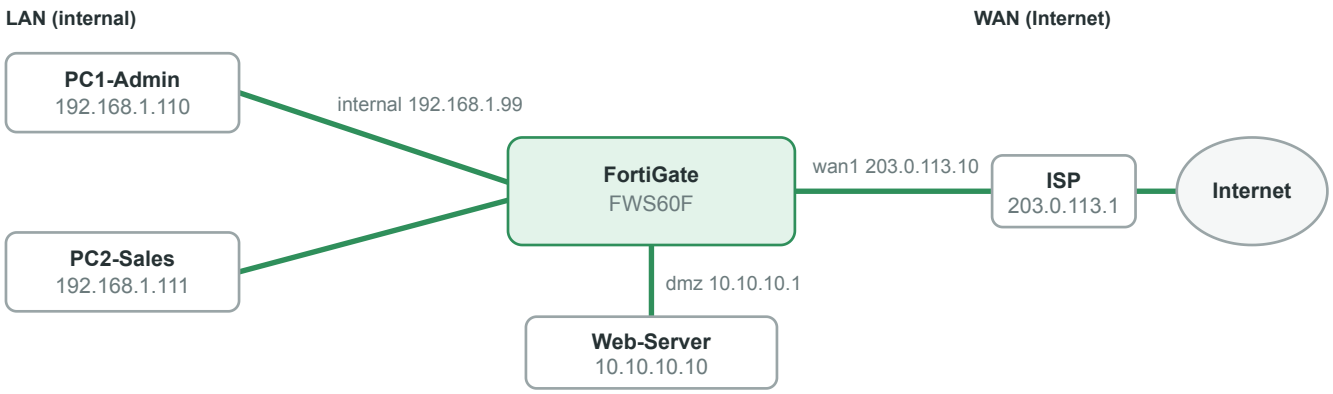
YOU WILL LEARN

- SSL-VPN settings
- portals
- ssl.root policy

BEFORE YOU START

Finish Class 10 first.

The lab network



The goal

THE IDEA

Remote staff need FortiClient access to the office PCs. Create a local user, a group, SSL-VPN settings on wan1 and the ssl.root policy, then connect Remote-Laptop.

How to start

1. Open FWallSim and log in.
2. Click **Labs** in the header.
3. Find **SSL-VPN for remote workers** and click **Start**.
4. The Lab Guide panel opens. Each task below gets a green tick as soon as it works.

Tasks

1. **SSL-VPN listens on wan1 on a port other than the admin HTTPS port**

   VPN > SSL-VPN Settings: Listen on wan1, port 10443.

2. **A local user is in a user group**

   User & Authentication > User Definition > Create New (Local User, with a password). Then User Groups > Create New (Firewall) and add the user as a member.

3. **An authentication rule maps that group to a tunnel-mode portal**

   VPN > SSL-VPN Settings > Authentication/Portal Mapping > Create New: your group, portal tunnel-access. That portal uses split tunneling, so only office traffic goes through the VPN and the laptop reaches the internet directly.

4. **A policy ssl.root → internal allows that group**

   Firewall Policy > Create New: Incoming SSL-VPN tunnel interface (ssl.root), Outgoing internal, Source all + your group, Destination the office LAN, Service ALL, ACCEPT.

5. **Remote-Laptop is connected with FortiClient**

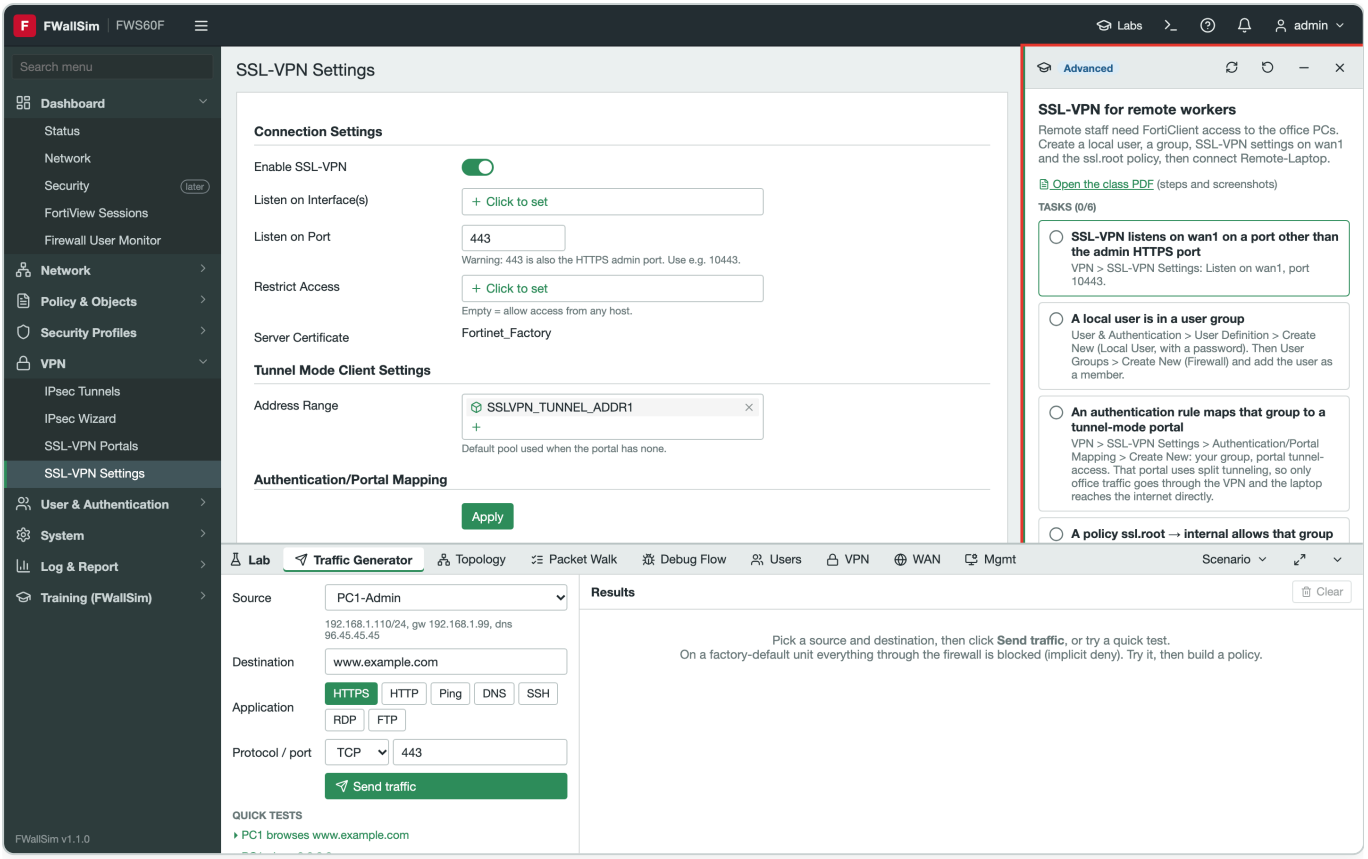
   Lab > VPN: connect Remote-Laptop to 203.0.113.10 on your SSL-VPN port with the local user.

6. **Remote-Laptop can RDP to PC1 through the tunnel**

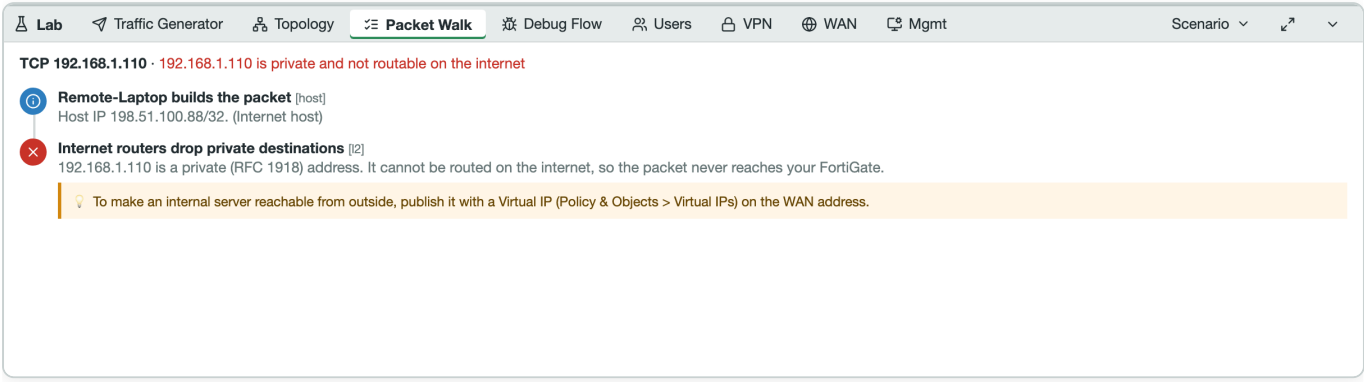
   Traffic Generator: click the quick test "Remote-Laptop RDP to PC1" (Remote-Laptop → 192.168.1.110, RDP TCP/3389).

Screenshots

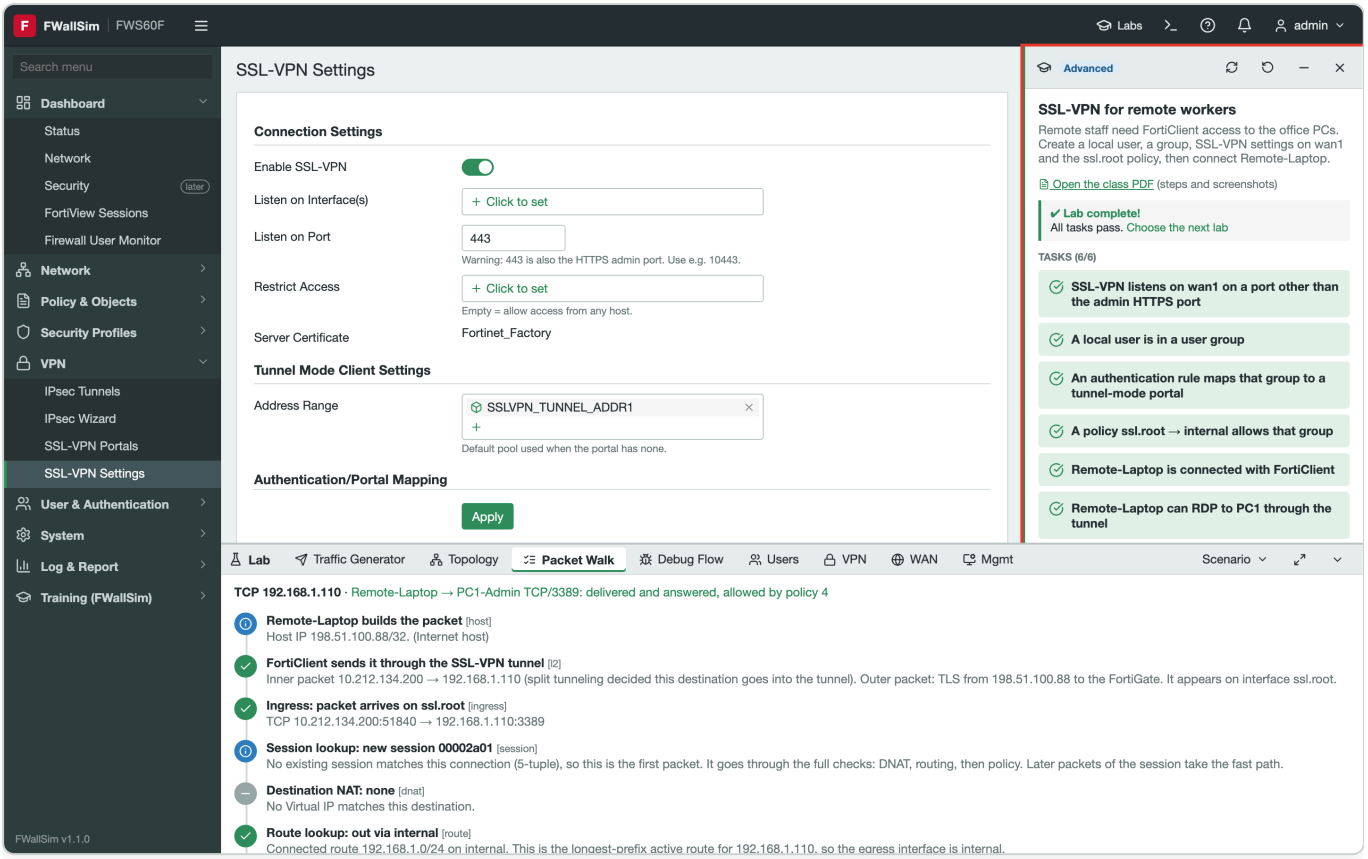
Numbered red markers show where to click or what to look at.



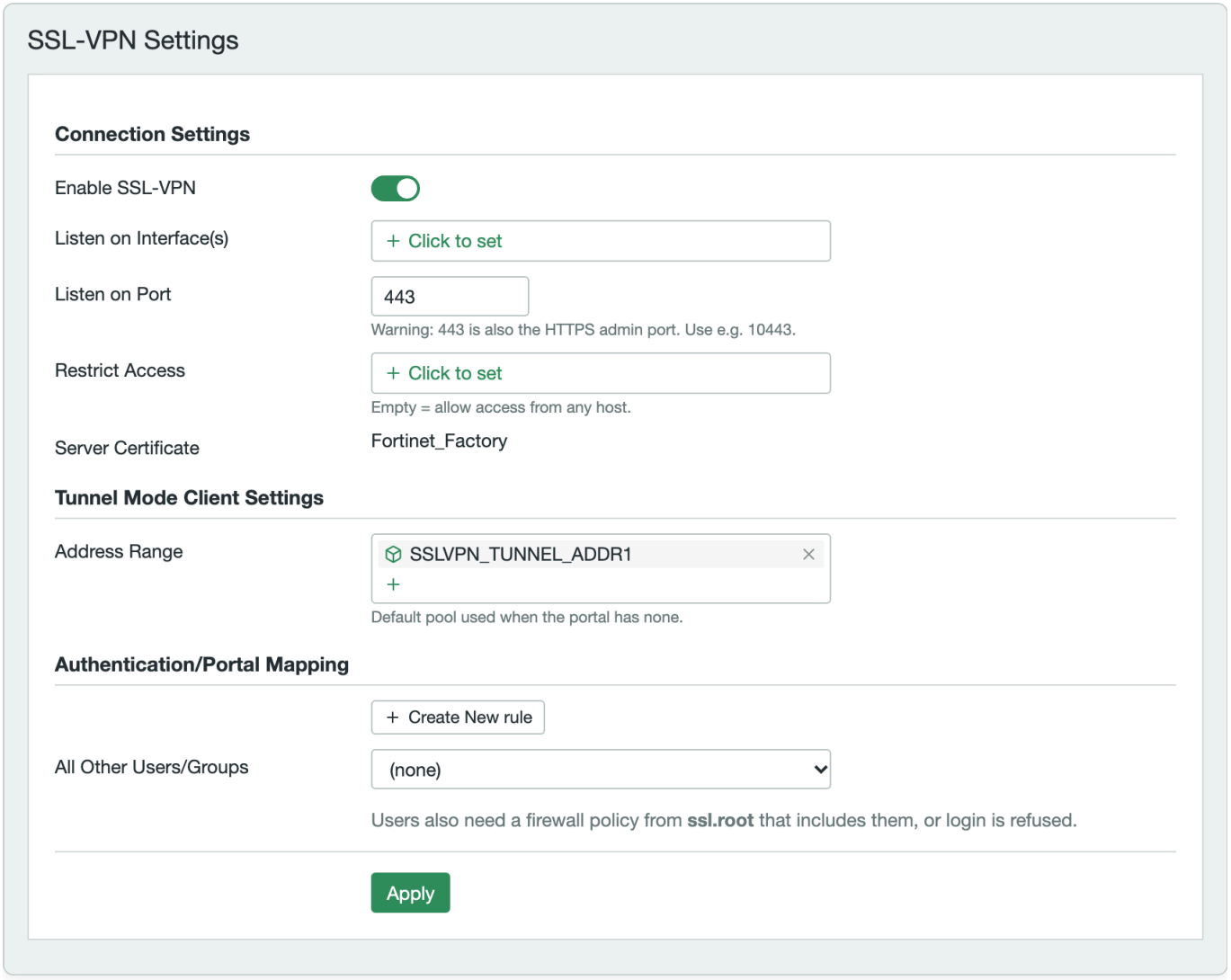
Lab Started



Packet Walk Before


FWallSim | FWS60F
Labs
admin
Search menu
Dashboard
Status
Network
Security
later
FortiView Sessions
Firewall User Monitor
Network
Policy & Objects
Security Profiles
VPN
IPsec Tunnels
IPsec Wizard
SSL-VPN Portals
SSL-VPN Settings
User & Authentication
System
Log & Report
Training (FWallSim)
FWallSim v1.1.0
SSL-VPN Settings
Connection Settings
Enable SSL-VPN
Listen on Interface(s)
+ Click to set
Listen on Port
443
Warning: 443 is also the HTTPS admin port. Use e.g. 10443.
Restrict Access
+ Click to set
Empty = allow access from any host.
Server Certificate
Fortinet_Factory
Tunnel Mode Client Settings
Address Range
SSLVPN_TUNNEL_ADDR1
Default pool used when the portal has none.
Authentication/Portal Mapping
Apply
Advanced
SSL-VPN for remote workers
Remote staff need FortiClient access to the office PCs. Create a local user, a group, SSL-VPN settings on wan1 and the ssl.root policy, then connect Remote-Laptop.
Open the class PDF (steps and screenshots)
✔ Lab complete!
All tasks pass. Choose the next lab
TASKS (6/6)
SSL-VPN listens on wan1 on a port other than the admin HTTPS port
A local user is in a user group
An authentication rule maps that group to a tunnel-mode portal
A policy ssl.root → internal allows that group
Remote-Laptop is connected with FortiClient
Remote-Laptop can RDP to PC1 through the tunnel
Lab
Traffic Generator
Topology
Packet Walk
Debug Flow
Users
VPN
WAN
Mgmt
Scenario
TCP 192.168.1.110 · Remote-Laptop → PC1-Admin TCP/3389: delivered and answered, allowed by policy 4
Remote-Laptop builds the packet [host]
Host IP 198.51.100.88/32. (Internet host)
FortiClient sends it through the SSL-VPN tunnel [l2]
Inner packet 10.212.134.200 → 192.168.1.110 (split tunneling decided this destination goes into the tunnel). Outer packet: TLS from 198.51.100.88 to the FortiGate. It appears on interface ssl.root.
Ingress: packet arrives on ssl.root [ingress]
TCP 10.212.134.200:51840 → 192.168.1.110:3389
Session lookup: new session 00002a01 [session]
No existing session matches this connection (5-tuple), so this is the first packet. It goes through the full checks: DNAT, routing, then policy. Later packets of the session take the fast path.
Destination NAT: none [dnat]
No Virtual IP matches this destination.
Route lookup: out via internal [route]
Connected route 192.168.1.0/24 on internal. This is the longest-prefix active route for 192.168.1.110, so the egress interface is internal.

Lab Solved



SSL-VPN Settings

Firewall Policy

+ Create New | Edit | Delete | Policy Lookup | Interface Pair View | By Sequence

Search

Name	ID	From	To	Source	Destination	Schedule	Service	Actic
internal → wan1 (1)								
LAN-to-Internet	1	internal	wan1	LAN-Subnet	all	always	ALL	✓ A
wan1 → dmz (1)								
Web-Server-Inbound	2	wan1	dmz	all	VIP-WebServer-HTTP	always	HTTP	✓ A
dmz → wan1 (1)								
DMZ-to-Internet	3	dmz	wan1	WebServer-DMZ	all	always	DNS HTTP HTTPS	✓ A
ssl.root → internal (1)								
ssl	4	ssl.root	internal	all VPN	LANnet	always	ALL	✓ A
Implicit (1)								
Implicit Deny	0	any	any	all	all	always	ALL	⊘ D

Policies are evaluated from top to bottom and the **first match wins**. Drag rows to reorder them. Right-click for more actions (insert, clone, enable/disable, logs). Right-click the Implicit Deny row to log violation traffic.

Firewall Policy

Lab | Traffic Generator | Topology | **Packet Walk** | Debug Flow | Users | VPN | WAN | Mgmt | Scenario

TCP 192.168.1.110 · Remote-Laptop → PC1-Admin TCP/3389: delivered and answered, allowed by policy 4

Remote-Laptop builds the packet [host]
Host IP 198.51.100.88/32. (Internet host)

FortiClient sends it through the SSL-VPN tunnel [l2]
Inner packet 10.212.134.200 → 192.168.1.110 (split tunneling decided this destination goes into the tunnel). Outer packet: TLS from 198.51.100.88 to the FortiGate. It appears on interface ssl.root.

Ingress: packet arrives on ssl.root [ingress]
TCP 10.212.134.200:50007 → 192.168.1.110:3389

Session lookup: new session 00002a02 [session]
No existing session matches this connection (5-tuple), so this is the first packet. It goes through the full checks: DNAT, routing, then policy. Later packets of the session take the fast path.

Destination NAT: none [dnat]
No Virtual IP matches this destination.

Route lookup: out via internal [route]
Connected route 192.168.1.0/24 on internal. This is the longest-prefix active route for 192.168.1.110, so the egress interface is internal.

Packet Walk After

Next: Class 12: SD-WAN with two ISPs